

NPRB2301: Estimation of shark body size from eye lenses (IN PROCESS)

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Objective

The purpose of this analysis is to estimate body size of Pacific sleeper sharks and Pacific spiny dogfish from images of sequentially peeled eye lenses, based on the presumed linear relationship between eye lens diameter and total body length. The estimation of body size at the time of formation of each lens layer analyzed for delta C-14 and stable isotope composition is intended to facilitate estimation of growth rates and life history reconstruction of individual sharks.

Introduction

Vertebrate eye lenses grow continuously throughout life, adding sequential layers around an embryonic core over time. Other researchers have demonstrated linear relationships between

the fish eye lens diameter and total length, including for the elasmobranchs porbeagle shark *Lamna nasus*, Atlantic spiny dogfish *Squalus acanthias*, thorny skate *Amblyraja radiata*, round ray *Rajella fyllae*, and leopard sharks *Triakis semifasciata* (Quaek-Davies et al. 2018; Leifsdottir and Campana 2023; Kuntz et al. 2025). In this subanalysis, we describe the relationships of whole lens diameter and the diameter of the hardened portion of the lens (i.e., post-apoptotic layers) to the total length of Pacific sleeper sharks *Somniosus pacificus* and Pacific spiny dogfish *Squalus suckleyi* to estimate the total length at the time of formation of each peeled eye layer in order to link delta C-14 and stable isotope values to body size.

Methods

Imaging and measurements

Eye lenses were dissected from frozen eye globes and photographed against a 10 x 10 mm grid background to spatially calibrate images for measurements (Figure 1). For most individuals, a photograph was taken of the whole, intact lens and of the lens following each sequential peel to a small central nucleus. Eye lenses differed in appearance. While the majority of whole lenses were spherical, a few were oblong in shape; many appeared firmer and more opaque, whereas other lenses were surrounded by a clear jelly layer (Figure 2). Measurements were generally taken through the equator at the widest point of the solid lens, unless the capsule was clearly beginning to delaminate from the rest of the lens. In those cases, the lens was measured where it appeared most spherical (Figure 3). Some lenses were too irregular to measure. The condition of the lens was noted when taking measurements. All measurements were made using ImagePro v.10-11 (Media Cybernetics) and converted from pixels to mm in R statistical computing software. Images of samples with observations outside the 99% confidence limits of the regressions of the whole lens diameter (LD_w) or first hardened lens diameter (LD_h) on the total length at capture (TL_c) were re-examined to identify potential measurement errors.

Comparison of capsule diameter to the diameter of the first hard layer

The outermost layer of the whole lens is termed the “capsule”. Just medial of the capsule is a layer of cuboidal epithelial cells, followed by the pre-apoptotic layers, and then the post-apoptotic hardened layers which are composed of crystalline proteins (Kuntz et al. 2025, Quaek-Davies et al. 2018). Studies have suggested that the ratio of the whole lens diameter (LD_w) and the first hardened layer (LD_h) is consistent across the life of the fish, regardless of species (Leifsdottir and Campana 2023). A regression of the LD_h against the LD_w was conducted by species, and a t-test was performed to determine if the slopes of the lines were significantly different from 1. A slope of 1 would be expected if the LD_w and LD_h increased equally as the eye grows. Similarly, we also examined the relationships of the $LD_h:LD_w$ to total body length at the time of capture (TL_c) and the $LD_h:LD_w$ to LD_w , but tested if those

slopes are significantly different from zero. In these cases a slope of zero would be expected if the ratio of $LD_h:LD_w$ was conserved as the eye or fish grows.

Relationships of LD_w and LD_h to TL

Measurement error in TL_c is a known source of uncertainty for this study. Samples were collected from many platforms, likely resulting in inconsistencies in total length measurements. Potential sources of variation include whether the tape measure was placed flat on the deck or over the curve of the body, inability to lay the shark flat due to limited deck space for laying out large specimen, and placement of the caudal fin (in a natural position or flexed). It is not possible to account for this uncertainty and therefore, total length is assumed measured without error for all analyses.

The relationship between LD_w and TL_c was described using separate linear models for each species. The equation followed the form:

$$LD_w = a_i + TL_c * b_i$$

where i is the species. The slope and intercept parameters are b and a , respectively.

Similarly, the relationship between LD_h and TL_c was described using the same equation

$$LD_h = c_i + TL_c * d_i$$

but the slope and intercept parameters were c and d , respectively. [NOTE: due to coding, these parameters are still listed as a and b for both models and will be fixed in final editing. See Table 2, which denotes them by LD_w and LD_h]

Eye lens diameter at age-0

Kuntz et al. (in review) demonstrate for leopard shark that the the diameter at age-0 can be identified by a significant peak or decline in the stable isotope values occurring after parturition, when the pups transition from maternal nourishment to self-feeding. A generalized additive model (GAM) was fit to each individual's d13C and d15N data. The first derivative of the d13C models, or when the slope of the model changed from negative to positive, was used to identify the minimum for each individual. A mean layer diameter at the model minimum across individuals is assumed to be a proxy for the hardened lens diameter at birth (with 95% confidence intervals).

Similarly, the first peak of the d15N should also correspond to birth.

Estimating fish length at the time of layer formation

The above equation can be rewritten to estimate the total length at the time of layer formation (TL_t) following Kuntz et al. (2025):

$$TL_t = (LD_t - a_{i,j})/b_{i,j}$$

where LD_t is the diameter of the lens following each sequential peel and the $a_{i,j}$ and $b_{i,j}$ parameters are from Table 2.

Two estimates of TL_t were estimated for each animal, using the LD_w vs TL and LD_h vs TL model parameters reported in Table 2. Uncertainty was estimated based on random draws of the linear regression parameters ($n = 1,000$, normal distribution around the parameter estimate) and the 2.5th and 97.5th quantiles of the resultant TL_t estimates.

Results

Comparison of capsule diameter to the diameter of the first hard layer

Paired LD_w and LD_h were available from 40 Pacific sleeper sharks and 8 Pacific spiny dogfish with total length recorded at time of capture. The slope of the regression line between LD_h and LD_w was significantly different from 1 ($p < 0.05$) for Pacific Sleeper Shark, but not Pacific Spiny Dogfish (Table 1, Model = “LDh vs LDw”, Figure 4). The slopes of the regression lines of the ratio of $LD_h:LD_w$ relative to the TL_c and relative to the LD_w were significantly different from zero for Pacific sleeper sharks, but not for Pacific spiny dogfish (Table 1, Model = “ratio vs TLc” and “ratio vs LDw”, respectively, Figure 5).

Relationship of LD_w and LD_h to TL

Paired LD_w and TL_c data were available from 65 Pacific sleeper sharks and 21 spiny dogfish, while paired LD_h and TL_c data were available from 62 Pacific sleeper sharks and 23 spiny dogfish (Table 2, Figure 6). The coefficients of determination (R^2) from the regressions of both of the LD_w and LD_h to TL_c for Pacific sleeper shark were > 0.7343 , suggesting that the linear regression explained most of the variability in the data. The R^2 of the regressions of both of the LD_w and LD_h to TL_c for Spiny Dogfish were > 0.9382 , suggesting that the linear regression explained most of the variability in the data.

Eye lens diameter at Age-0

The GAM fits for each individual by species are in Figure 7. The mean of the first derivative, or localized minima, across all individuals in a species resulted in estimated diameter at age-0 of 5.7518 (5.2704-5.2704, 95% CI) and 5.1843 (4.8472-4.8472, 95% CI) for Pacific sleeper shark and spiny dogfish, respectively. Spiny dogfish are known to be highly mobile, and may move through many regions during the 18-22 month gestation period, which may explain the high variability in the $\delta^{13}\text{C}$ curves. Conversely, while it is unknown exactly where adult Pacific sleeper sharks reside, it is believed that adults of the species generally inhabit the abyssal plain or other deep offshore environments. One current hypothesis is that pregnant females move to nearer to shore to pup and the pups move to shallow nursery areas (e.g., Bering Sea). The consistent valley in $\delta^{13}\text{C}$ across individuals suggests a similar movement pattern at the same size, possibly from the abyssal environment while in utero to a nearshore nursery habitat after pupping.

The $\delta^{15}\text{N}$ data are more variable across individuals in both species. This could be explained by the large size at birth and the generalist/opportunistic feeding strategy following birth. Even though pups are more gape-limited than adults, pups of both species could potentially feed at some of the same trophic levels as the adults. The initial peak of $\delta^{15}\text{N}$ is difficult to identify in most individuals and may be less informative for inferring lens diameter at birth.

Estimating fish length at the time of layer formation

The TL_t was estimated for each layer diameter based on the linear regression, with uncertainty (Figure 8). The TL_t estimates derived from the regression of the LD_w were smaller for both species and resulted in many negative TL_t estimates and many TL_t well below the hypothesized size at birth.

Discussion

This work illustrates that the type and composition of the layer diameters being used to predict length need to be considered. The ratio of the whole lens diameter to the first hardened layers (or diameter at the first post-apoptotic layer) increases with the size of the eye. It is unclear if this is a function of time or size, or both. The LD_w includes the pre-apoptotic layers and the capsule surrounding the whole lens. The process of apoptosis may occur at different rates throughout the growth trajectory of an animal, resulting in the significant change in the relationship between the LD_h and LD_w . It would be incorrect to make further predictions without considering the different biological processes between the LD_h and LD_w .

The majority of the layer diameter data for this study is from the post-apoptotic layers. For consistency, we recommend using the linear model between the LD_h and TL_c to predict TL_t . The LD_w is representative of the TL_c , not to post-apoptotic processes.

The stable isotope data does not appear to be informative for estimating lens diameter at age-0, but further discussions and interpretations may change this.

Tables

Table 1: Linear regression model results with the slope, intercept, 95% confidence intervals, standard error, Rsquared, degrees of freedom, t-test p-value at alpha = 0.05, the model and the hypothesis that was tested.

Species	Slope	Lower	Upper	StErr	Rsqr	DF	Pval	Model	Hypothesis
SD	0.2813	-0.2354	0.7980	0.2112	0.2282	6	0.0144	LDh vs LDw	mu = 1
PSS	0.6946	0.5907	0.7985	0.0513	0.8282	38	0.0000	LDh vs LDw	mu = 1
SD	-0.0255	-0.0676	0.0167	0.0172	0.2673	6	0.1895	ratio vs LDw	mu = 0
PSS	0.0187	0.0093	0.0282	0.0046	0.2999	38	0.0003	ratio vs LDw	mu = 0
SD	-0.0001	-0.0006	0.0004	0.0002	0.0316	6	0.6734	ratio vs TL	mu = 0
PSS	0.0001	0.0000	0.0001	0.0000	0.3498	38	0.0001	ratio vs TL	mu = 0

Table 2: Linear regression statistics of the whole lens diameter (LD_w) and the first hardened layer (LD_h) to total length at capture (TL_c) for each species (i), with 95% confidence intervals, standard error, Rsquared, degrees of freedom, t-test p-value at alpha = 0.05, and the model (j).

Species	Parameter	Value	Lower	Upper	StErr	Rsqr	DF	Pval	Model
SD	a	3.6747	2.9014	4.4479	0.3694	0.9382	19	0e+00	LDw vs TL
SD	b	0.0091	0.0080	0.0103	0.0005	0.9382	19	0e+00	LDw vs TL
PSS	a	5.5507	4.8192	6.2823	0.3661	0.7467	63	0e+00	LDw vs TL
PSS	b	0.0025	0.0021	0.0029	0.0002	0.7467	63	0e+00	LDw vs TL
SD	a	1.6866	1.2524	2.1208	0.2088	0.9545	21	0e+00	LDh vs TL
SD	b	0.0063	0.0057	0.0069	0.0003	0.9545	21	0e+00	LDh vs TL
PSS	a	0.9278	0.4030	1.4526	0.2624	0.7343	60	8e-04	LDh vs TL
PSS	b	0.0019	0.0016	0.0021	0.0001	0.7343	60	0e+00	LDh vs TL

Figures

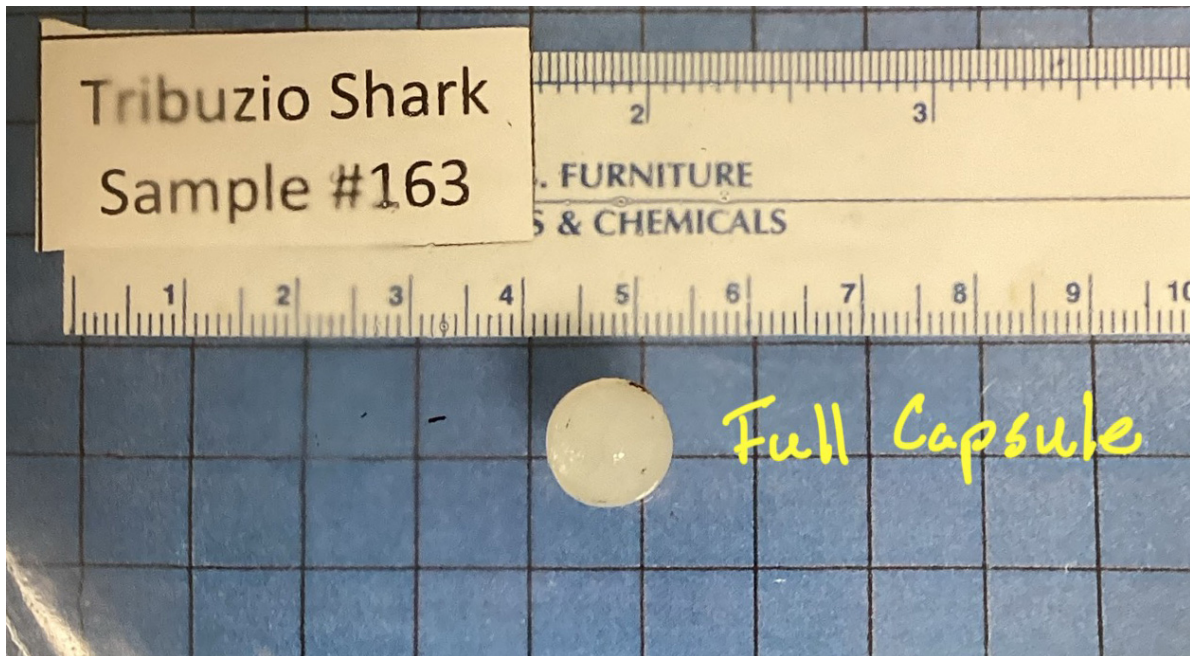


Figure 1: Image showing a whole, intact eye lens against gridded 10 x 10 mm background.

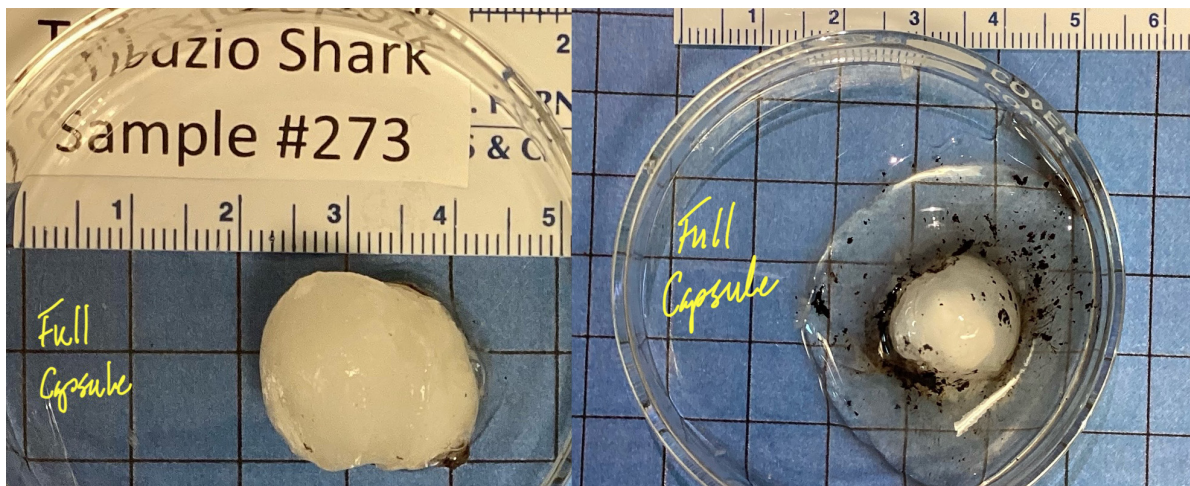


Figure 2: Eye lenses from 2 sharks of about the same length (left = 371 cm TL, right = 403 cm TL). Note the oblong shape of the lens on left and jelly layer surrounding the lens on right.

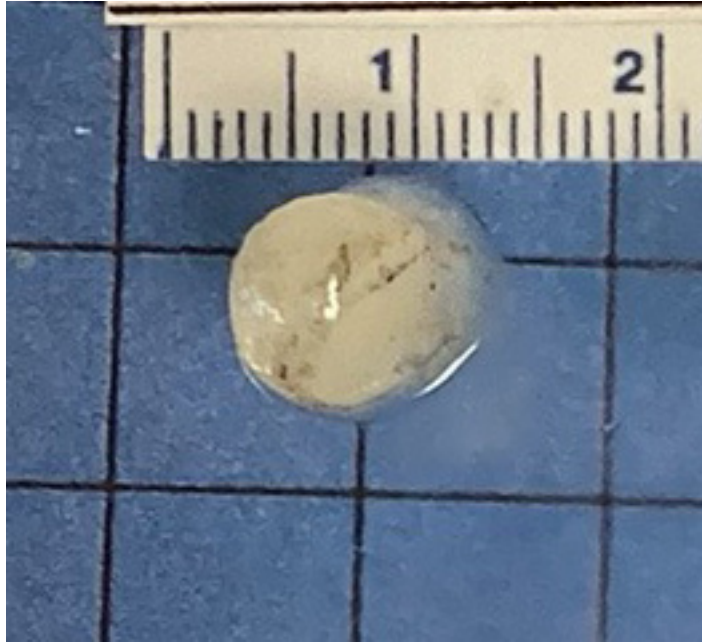


Figure 3: Outer capsule starting to separate from lens. Lens was measured where it is most spherical.

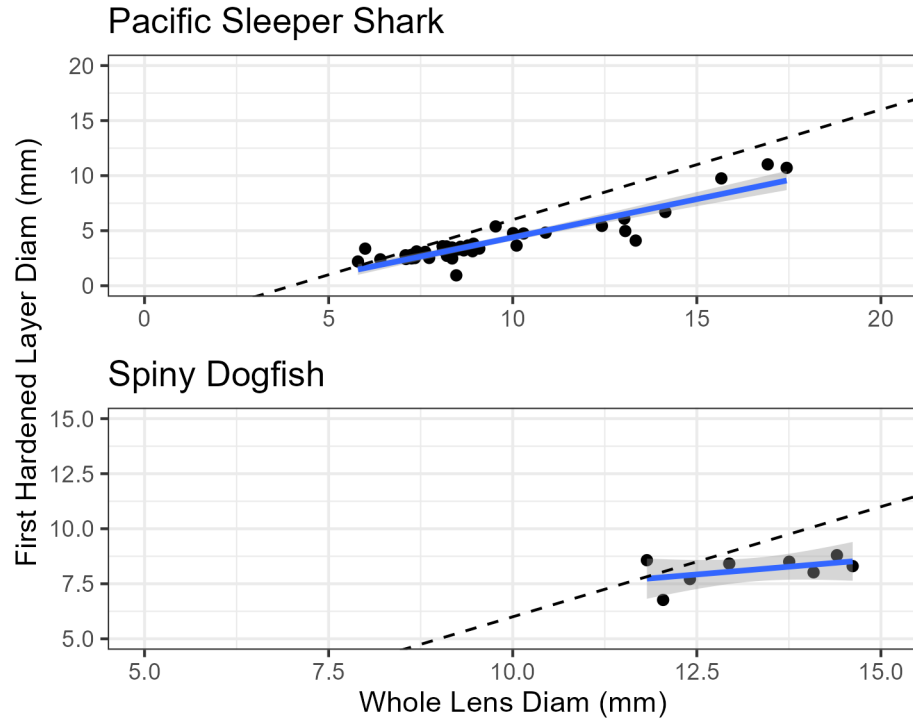


Figure 4: Linear regressions of the diameter (in millimeters) of the whole lens (LD_w) and the first hardened layer (LD_h). The shaded region represents the 95% confidence interval. The dashed line is the expected 1:1 replacement line, null hypothesis $\mu = 1$.

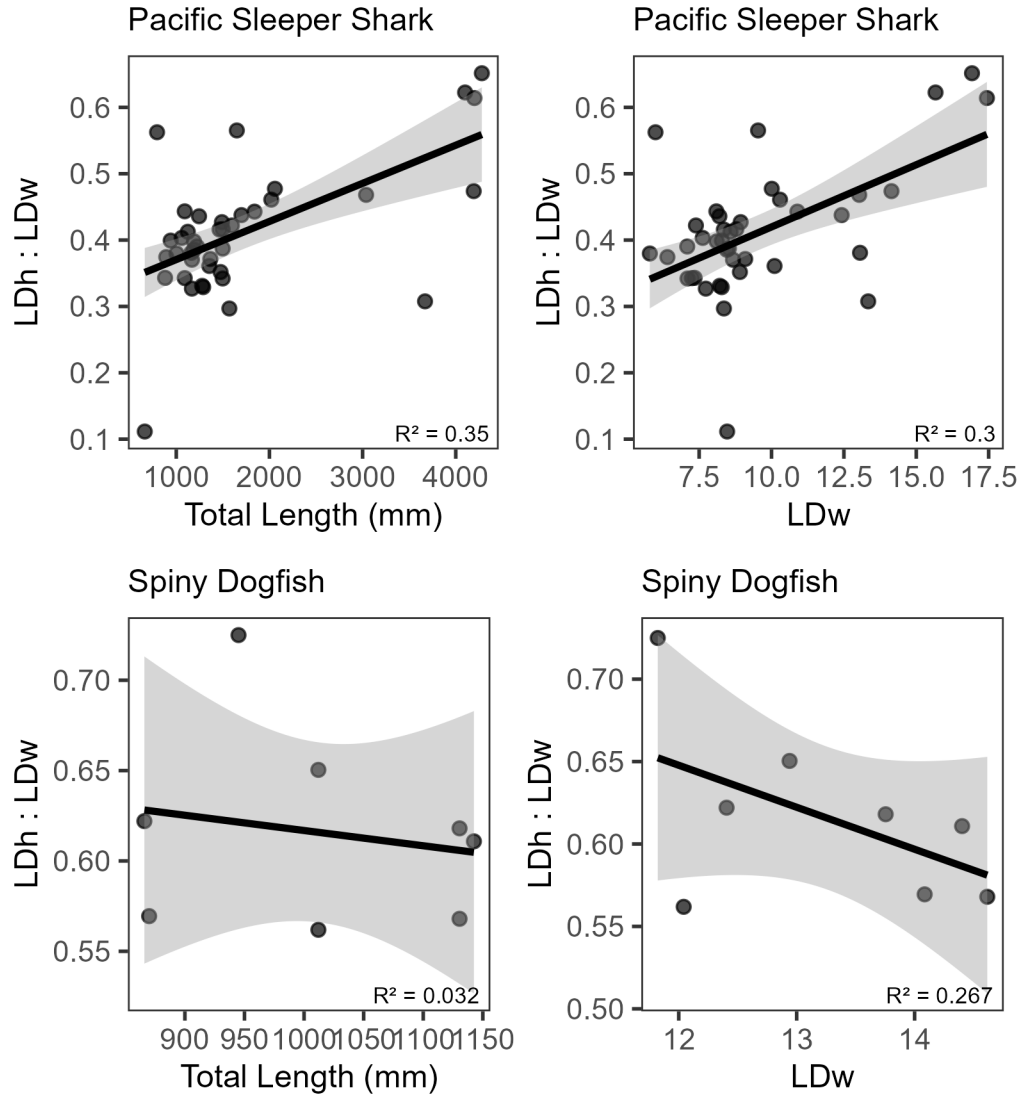


Figure 5: Linear regressions of the ratio of the diameter of the first hardened layer (LD_h) to the diameter of the whole lens (LD_w) relative to the total length at capture (TL_c) and to the LD_w . Shaded region represents the 95% confidence interval.

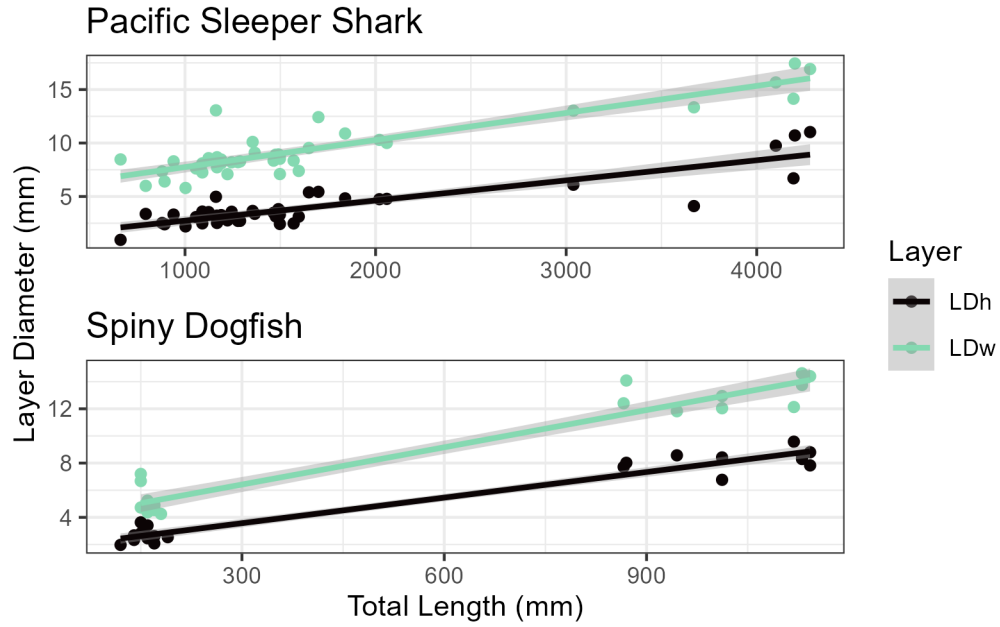


Figure 6: Linear regressions of the LD_h and the LD_w to the total length at capture. Shaded region represents the 95% confidence interval.

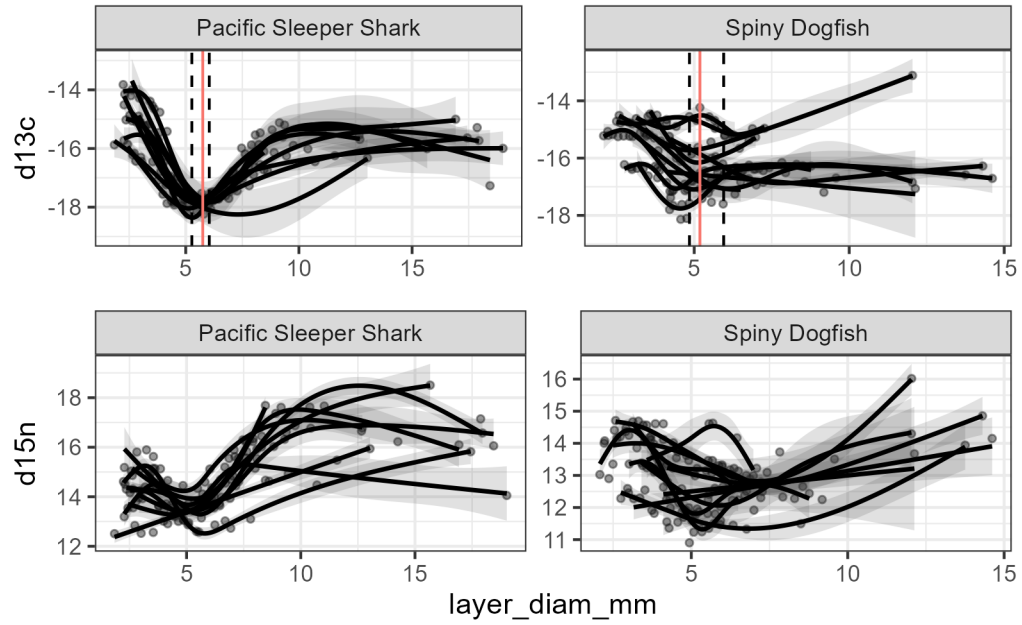


Figure 7: Bulk $d_{13}C$ and $d_{15}N$ data with individual fitted GAMs for each animal. Solid vertical lines are the mean layer diameter of the model minimum $d_{13}C$, Dashed lines are the 95% confidence interval

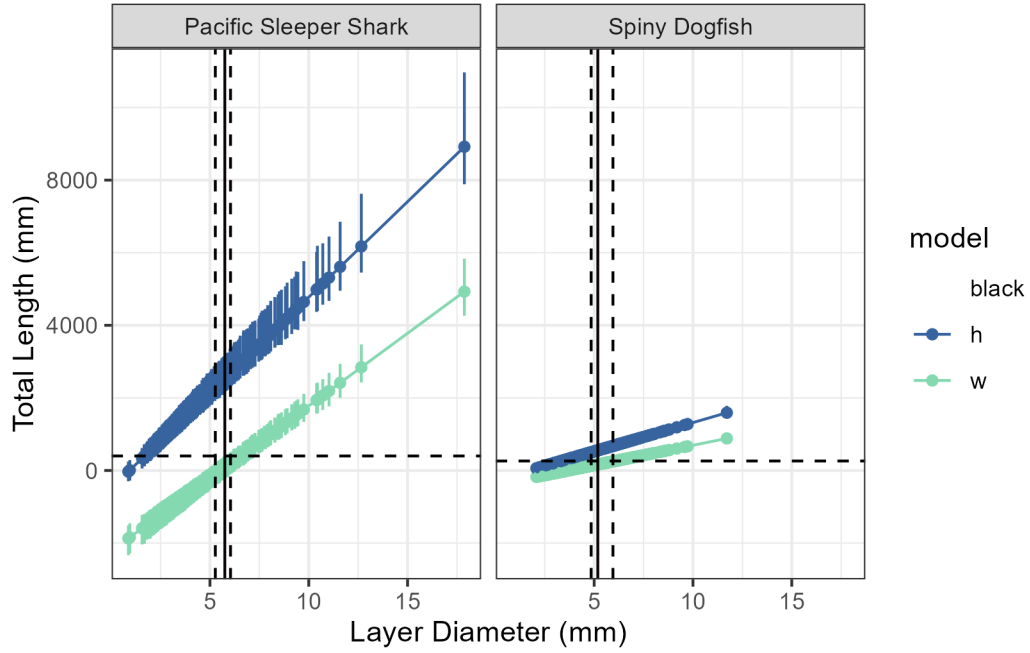


Figure 8: Predicted total lengths (in mm) based on the whole lens diameters (LD_w) and first hardenend layer (LD_h) linear models. The horizontal dashed line is the estimated size at birth from literature. The vertical solid black line is the estimated diameter at birth based on the d13C data, with 95% confidence intervals in dashed lines.